

End-to-End SQL Data Analysis Project | Automobile Dataset | MySQL Portfolio Project

<https://www.kaggle.com/>

<https://www.kaggle.com/datasets/tawfikelmetwalli/automobile-dataset?>

Here is the breakdown of dataset columns

column name	Description
Name	Name of the car
mpg	Number of engine cylinders
displacement	engine size in cubic inches
horsepower	engine output
weight	weight of the car
acceleration	Time taken to go from 0 to 60kmph

model year	year of the car(last two digits)
origin	country of origin

1. Get all cars with mileage (mpg) above 30

select name,mpg

from automobile

where mpg>30;

name	mpg
datsum b-210	32
honda accord cvcc	32
renault 5 gtl	36
datsum f-10 hatch...	34
chevrolet chevette	31
dodge colt m/m	34
volkswagen dasher	31
volkswagen rabbit ...	43
ford fiesta	36
mazda glc deluxe	33
datsum b210 gx	39
honda civic cvcc	36

2. Find average horsepower for each origin

select origin,avg(horsepower) from
automobile group by origin;

Result Grid		Filter Rows:	Export:
origin	avg(horsepower)		
usa	119.0490		
japan	79.8354		
europa	80.5588		

3. Count number of cars from each origin

```
select origin,count(*) from automobile
group by origin
```

Result Grid		Filter Rows:
origin	count(*)	
usa	245	
japan	79	
europa	68	

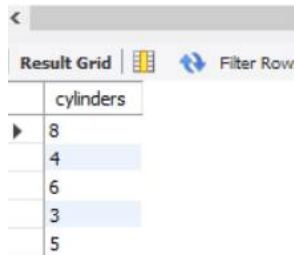
4. Find top 5 most fuel-efficient cars

```
select name,mpg
from automobile
order by mpg desc
limit 5;
```

Result Grid		Filter Rows:
name	mpg	
mazda glc	47	
honda civic 1500 gl	45	
vw rabbit c (diesel)	44	
vw pickup	44	
vw dasher (diesel)	43	

5. List distinct cylinder counts in the dataset

```
select distinct cylinders  
from automobiles;
```

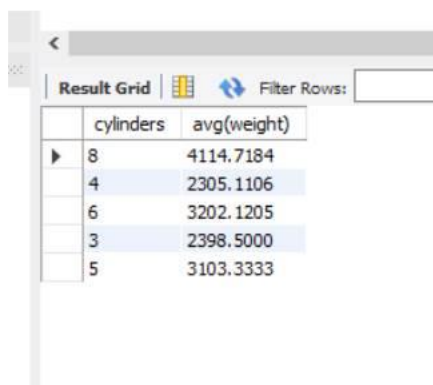


A screenshot of a database query result grid. The grid has a single column labeled 'cylinders'. The rows contain the values 8, 4, 6, 3, and 5. The row with '4' is highlighted in blue.

cylinders
8
4
6
3
5

6. Find average weight by number of cylinders

```
select cylinders, avg(weight)  
from automobile  
group by cylinders;
```



A screenshot of a database query result grid. The grid has two columns: 'cylinders' and 'avg(weight)'. The rows show the average weight for each cylinder count: 8 (4114.7184), 4 (2305.1106), 6 (3202.1205), 3 (2398.5000), and 5 (3103.3333). The row with 4 cylinders is highlighted in blue.

cylinders	avg(weight)
8	4114.7184
4	2305.1106
6	3202.1205
3	2398.5000
5	3103.3333

7. Find all cars with acceleration on below 10 seconds

```
select name, acceleration  
from automobile  
where acceleration < 10;
```

Result Grid			Filter Rows:
	name	acceleration	
▶	chevrolet impala	9	
	plymouth fury iii	8.5	
	amc ambassador dpl	8.5	
	plymouth 'cuda 340	8	
	chevrolet monte carlo	9.5	
	pontiac grand prix	9.5	

8. Get model years with the highest average mpg

```
select model_year,avg(mpg)
from automobile
group by model_year;
```

Result Grid			Filter Rows:
	model_year	avg(mpg)	
▶	70	17.6897	
	71	21.1111	
	72	18.7143	
	73	17.1000	
	74	22.7692	
	75	20.2667	
	76	21.7353	
	77	23.6786	
	78	24.1111	
	79	25.1724	
	80	33.7778	
	81	30.2143	

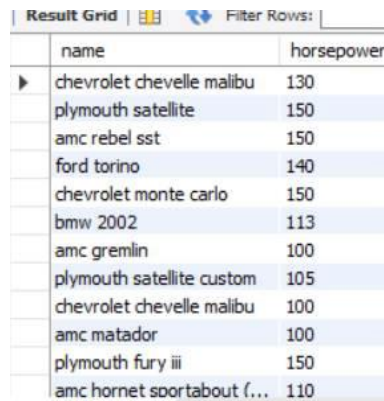
9. Find the heaviest car in the dataset

```
select name,weight
from automobile
order by weight desc
limit 1;
```

Result Grid			Filter Rows:
	name	weight	
▶	pontiac safari (sw)	5140	

10. List cars with horsepower between 100 and 150

```
select name, horsepower
from automobile
where horsepower between 100 and 150;
```



A screenshot of a database result grid titled 'Result Grid'. It shows a table with two columns: 'name' and 'horsepower'. The table contains 14 rows of data, with the first row highlighted in blue. The data is as follows:

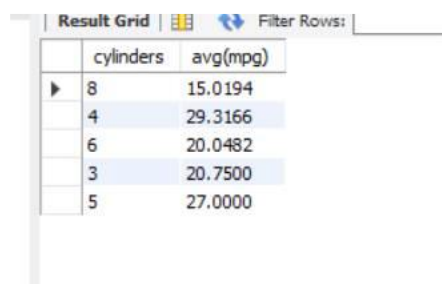
name	horsepower
chevrolet chevelle malibu	130
plymouth satellite	150
amc rebel sst	150
ford torino	140
chevrolet monte carlo	150
bmw 2002	113
amc gremlin	100
plymouth satellite custom	105
chevrolet chevelle malibu	100
amc matador	100
plymouth fury iii	150
amc hornet sportabout (...	110

11. Show the average mpg by number of cylinders

```
select cylinders, avg(mpg)
from automobile
group by cylinders;
```

12. Find all cars manufactured in 1975

```
select name, model_year
from automobile
where model_year=75;
```



A screenshot of a database result grid titled 'Result Grid'. It shows a table with two columns: 'cylinders' and 'avg(mpg)'. The table contains 5 rows of data, with the first row highlighted in blue. The data is as follows:

cylinders	avg(mpg)
8	15.0194
4	29.3166
6	20.0482
3	20.7500
5	27.0000

12. Find all cars manufactured in 1975

```
select name, model_year
from automobile
```

where model_year=75;

Result Grid   Filter Rows:		
	name	model_year
▶	plymouth valiant custom	75
	chevrolet nova	75
	mercury monarch	75
	ford maverick	75
	pontiac catalina	75
	chevrolet bel air	75
	plymouth grand fury	75
	ford ltd	75
	buick century	75
	chevrolet chevelle malibu	75
	amc matador	75
	plymouth fury	75

automobile 14 ✕